Experiment Name: 8-bit Division using Assembly Language

1. Objective

The main aim of this experiment is to divide two 8-bit numbers by using Assembly Language in the 8086 microprocessor. Through this experiment, we can learn how the DIV instruction works and how quotient and remainder are stored in registers.

2. Apparatus / Tools

- Personal Computer (PC)
- EMU8086 Emulator Software

3. Working Procedure

Steps:

1.The dividend is loaded into the AL register.
2. The divisor is loaded into the BL register.
3. The DIV instruction is used to perform division.
4. The quotient is stored in the AL register and remainder in AH register.

4. Program:

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE

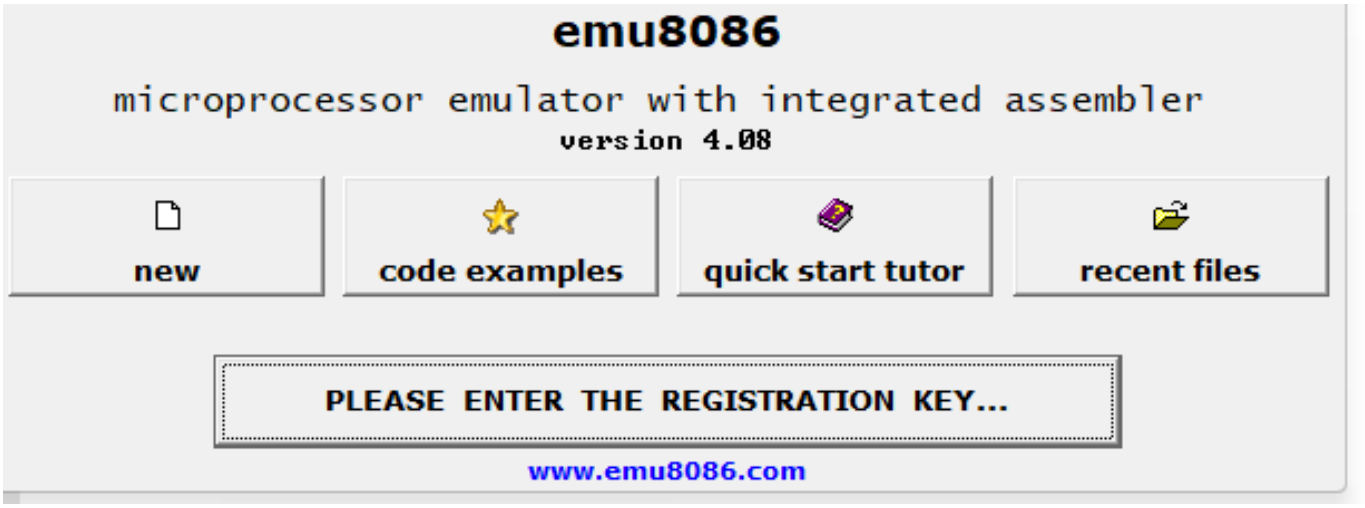
MAIN PROC

  MOV AL,48H
  MOV BL,08H
  DIV BL
  INT 21H

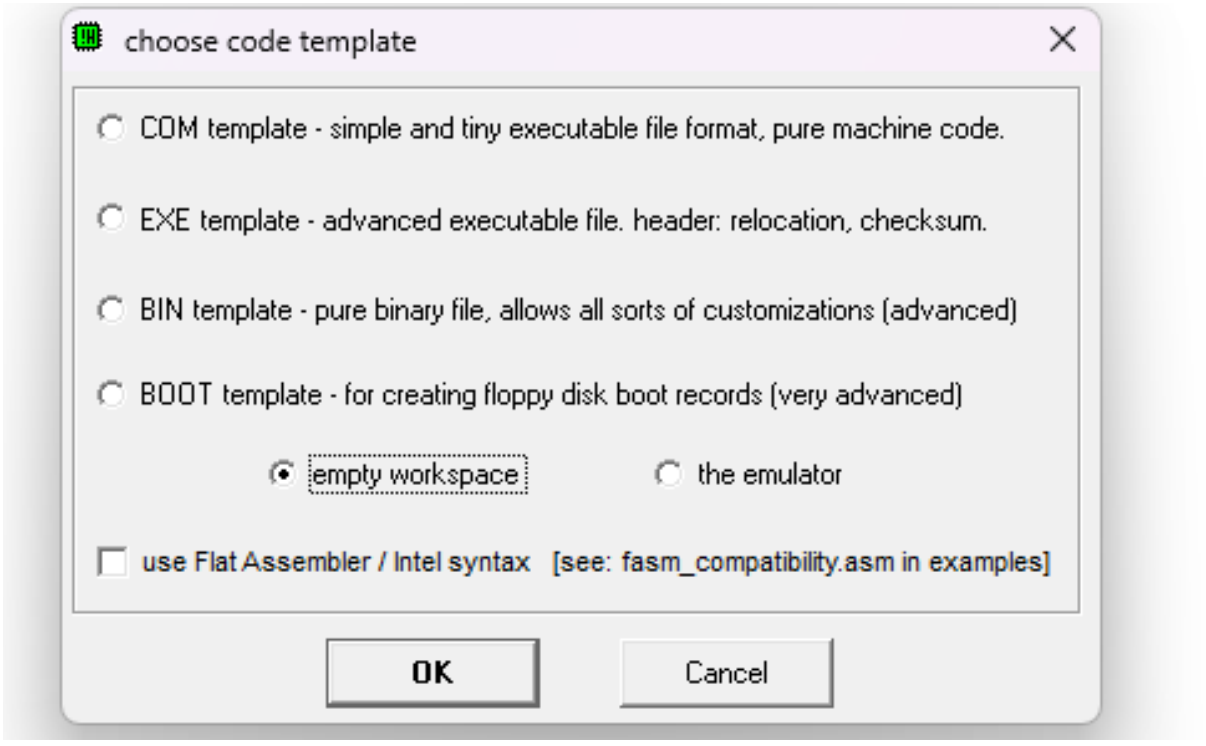
 MAIN ENDP
END MAIN
```

5.Operation Step

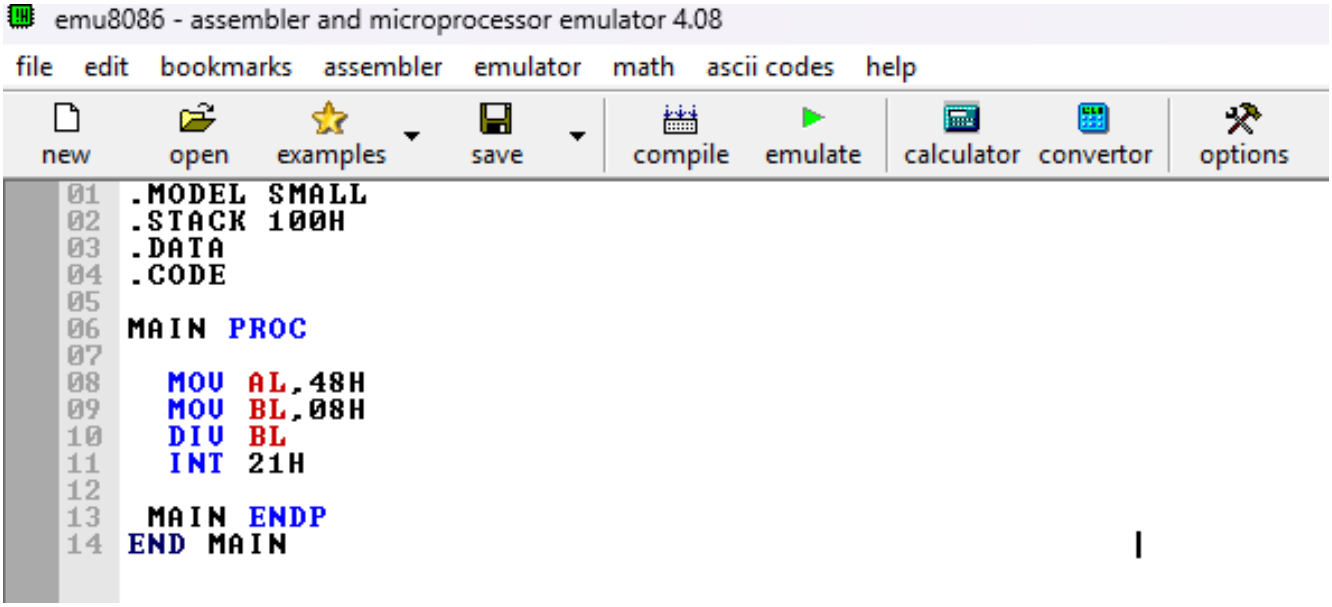
Step 1: Open EMU8086 and click on 'New' to start a project.



Step 2: Select “Empty Workspace”
After clicking on “New”, a new window will appear. From there, select “Empty Workspace” and click OK to proceed.

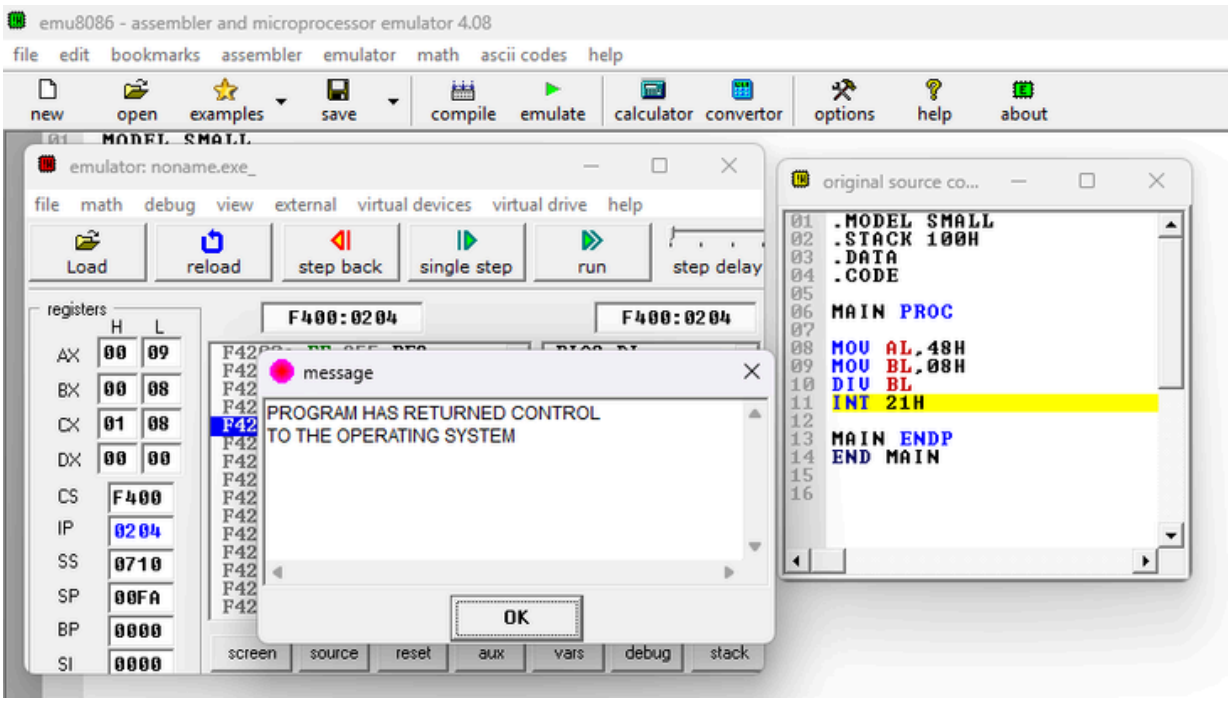


Step 3: Write the Program Code
After opening the workspace, write the assembly language program in the editor window.



5.Operation Result

Step 1 :Click on the Run/Execute option to run the program. The result of the addition will be shown in the register (AL).
Quotient = 09H
Remainder = 00H



6.Comment

The program executed successfully, and the 8-bit division was correctly performed with the quotient stored in the AL register and remainder stored in the AH register.

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